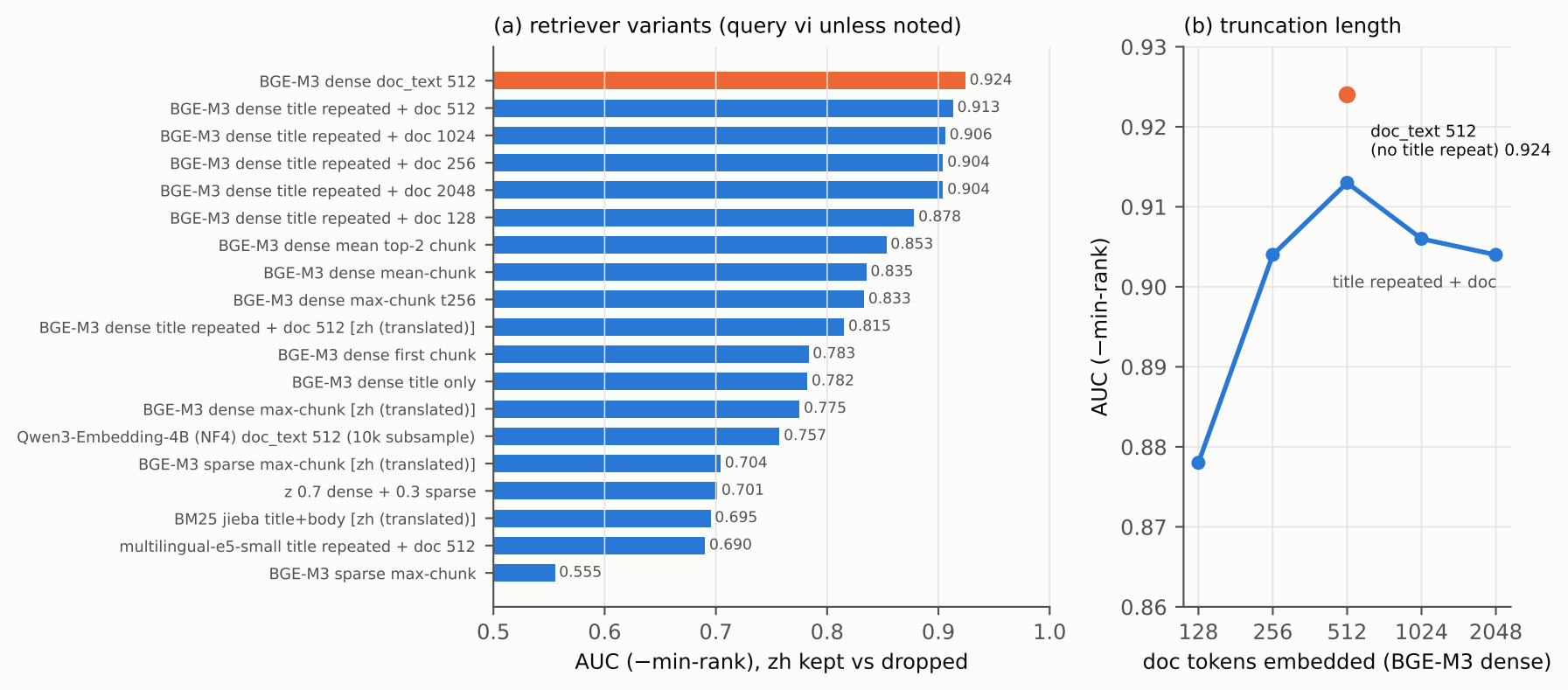
(a) retriever variants (query vi unless noted)
BGE-M3 dense doc_text 512
BGE-M3 dense title repeated + doc 512
BGE-M3 dense title repeated + doc 1024
BGE-M3 dense title repeated + doc 256
BGE-M3 dense title repeated + doc 2048
BGE-M3 dense title repeated + doc 128
BGE-M3 dense mean top-2 chunk
BGE-M3 dense mean-chunk
BGE-M3 dense max-chunk t256
BGE-M3 dense title repeated + doc 512 [zh (translated)]
BGE-M3 dense first chunk
BGE-M3 dense title only
BGE-M3 dense max-chunk [zh (translated)]
Qwen3-Embedding-4B (NF4) doc_text 512 (10k subsample)
BGE-M3 sparse max-chunk [zh (translated)]
z 0.7 dense + 0.3 sparse
BM25 jieba title+body [zh (translated)]
multilingual-e5-small title repeated + doc 512
BGE-M3 sparse max-chunk
0.924
0.913
0.906
0.904
0.904
0.878
0.853
0.835
0.833
0.815
0.783
0.782
0.775
0.757
0.704
0.701
0.695
0.690
0.555
0.5
0.6
0.7
0.8
0.9
1.0
AUC (−min-rank), zh kept vs dropped
(b) truncation length
0.93
0.92
0.91
0.90
0.89
0.88
0.87
0.86
AUC (−min-rank)
doc_text 512
(no title repeat) 0.924
title repeated + doc
128
256
512
1024
2048
doc tokens embedded (BGE-M3 dense)